

Power BI Assignment 2 — Power Query, TMDL & Date Intelligence

Objective

Practice transformation logic in Power Query, then create a professional calendar table and push a set of DAX measures into the model using TMDL view.

Dataset

Use Task_2_Power_Query_TMDL_Calendar_Data.xlsx. Main tables: FactOrders, DimEmployee and HolidaySeed. The raw status field intentionally contains inconsistent values such as blanks and N/A.

Assignment Requirements

1. Create a Custom Column that calculates line revenue from Units and UnitPrice, then compare it with the supplied Revenue column.
2. Create a Custom Column that calculates a margin/profit classification using Revenue, Cost and Profit.
3. Create at least 3 Conditional Columns: order status normalization, customer/order value band, and rating band.
4. Clean RawStatus so blanks and N/A become a meaningful standardized category.
5. Create a proper Calendar table covering the full FactOrders date range. Include Date, Year, Quarter, Month Number, Month Name, Year-Month, Week Number and Day Name.
6. Sort Month Name by Month Number and Year-Month by a numeric/date sort column. 7. Mark the Calendar table as the model Date table and create a one-to-many relationship to FactOrders.
8. Use TMDL view to create or push at least 8 measures into the model. Suggested measures: Total Revenue, Total Cost, Total Profit, Profit Margin %, Orders, Units, Average Order Value and Return Rate.
9. Create time-intelligence measures such as Revenue YTD, Revenue Previous Year and YoY Growth %.
10. Build a compact dashboard showing Revenue, Profit, Margin, Orders and YoY trend by Month and Department/Channel.

Required Deliverables

- PBIX/PBIP containing the transformed model.
- Power Query steps clearly showing Custom Column and Conditional Column logic.
- Calendar table with correct sorting and date relationship.
- TMDL-created measures visible in the semantic model.
- One dashboard page with KPI and trend validation.

Evaluation Checklist

Area	Expected evidence
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Power Query	Transformations are reproducible and performed before loading where appropriate.
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Calendar	Continuous date table, correct data types, sorting, and marked Date table.
TMDL	Measures are created through TMDL view and have clear names/formats.
DAX	Measures return sensible values and time intelligence works.
Visualization	Dashboard demonstrates the measures rather than merely listing them.

Submission note: Keep the model clean, use meaningful names, avoid unnecessary calculated columns, and document important modeling decisions.